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Research Project on the Topic:
Evaluation of a Reinforcement Learning Framework
in a Credit Scoring Problem

Submitted by the Student:

group #БПАД221, 4th year of study Baulin Evgeny

Approved by the Supervisor:

HSE, Big Data and Information Retrieval School Peter Lukianchenko

18.05.2026
Date

Signature

Lukianchenko P.P.
Initials

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Annotation

This thesis evaluates a reinforcement learning (RL) framework for weekly credit-scoring threshold optimization in a simulated retail lending portfolio. The core decision problem is not a static approval or rejection of individual loan applications but a sequential portfolio policy: each week the controller sets two acceptance thresholds—one for new clients and one for repeat clients—and then observes delayed consequences in the form of repayments, defaults, recoveries, capital usage, and realized profit over subsequent weeks.

The study employs a synthetic credit-portfolio simulator calibrated to published retail lending statistics and covering six market scenarios: stable conditions, gradual score drift, adverse stress, class-imbalance shifts, noisy measurements, and divergent segment dynamics. Six reinforcement learning algorithms—DQN, Double-DQN, A2C, PPO, SAC, and A3C—are trained and evaluated against five rule-based or static baselines (`profit_oriented`, `constraint_aware_weekly`, `risk_aware_weekly`, `static_threshold`, and `split_policy_static`). A controlled dimensionality experiment compares four observation representations (12D, 20D, 30D, and 50D features) while holding all other experimental factors fixed.

The main findings are as follows. The best overall controller is the rule-based `constraint_aware_weekly` baseline, which wins all six scenarios by cumulative reward and expected profit, strongly confirming Hypothesis H2 that RL does not achieve universal superiority. The best RL agent, Double-DQN, peaks at 12D. The 30D versus 12D improvement is not statistically significant (H1 partially supported), while 50D significantly underperforms 30D. Under full-profile adverse-stress conditions, the static threshold baseline significantly outperforms the 30D DQN policy, reversing the pattern observed at smaller training budgets (H3 rejected).

Limitations include the synthetic nature of the simulator, the designed reward function, the short training budget in the quick profile, and the episode-static threshold behavior of the best RL policies.

Key Words

Credit scoring — a quantitative method for evaluating the creditworthiness of a loan applicant by assigning a numerical score that reflects the estimated probability of default.

Acceptance threshold — the minimum credit score above which a loan application is approved by a lender’s portfolio policy. Adjusting the threshold alters the trade-off between approval volume and portfolio default risk.

Reinforcement learning (RL) — a branch of machine learning in which an agent learns a policy by interacting with an environment, receiving scalar rewards, and maximizing expected cumulative return over time.

Markov Decision Process (MDP) — a formal framework for sequential decision-making under uncertainty, defined by a state space, an action space, a transition function, and a reward function. Reinforcement learning problems are typically formulated as MDPs.

Reward shaping — the practice of augmenting or modifying the environmental reward signal to accelerate learning or encode domain-specific objectives such as risk constraints and capital-usage limits.

Delayed reward — a reward signal that arrives several time steps after the action that produced it. In credit scoring, delayed rewards arise because loan repayments and defaults occur weeks or months after the acceptance decision.

State representation — the observable summary of the environment’s current condition provided to the agent at each decision step. In this thesis it ranges from a compact 12-dimensional weekly dashboard to a 50-dimensional segmented history.

Portfolio simulation — the use of a synthetic or model-based environment to generate realistic sequences of loan applications, repayments, and defaults for the purpose of training and evaluating portfolio-management policies.

DQN (Deep Q-Network) — a model-free reinforcement learning algorithm that approximates the action-value function with a deep neural network, using experience replay and a target network to stabilize training.

Rule-based baseline — a policy that makes decisions according to a fixed or heuristic rule rather than learned parameters, used in this thesis to benchmark whether reinforcement learning agents achieve practically meaningful improvements over simpler alternatives.

Contents

1	Introduction	6
2	Literature Overview	7
2.1	Credit Scoring, Cut-off Optimisation, and Portfolio Risk	7
2.2	Reinforcement Learning for Sequential Financial Decisions	7
2.3	Simulation, Robustness Evaluation, and Distribution Shift	8
3	Problem Framing and Research Motivation	8
3.1	Why weekly threshold control is the research object	8
3.2	Why delayed outcomes and delayed rewards matter	9
3.3	Why separate thresholds for new and repeat clients are essential	9
4	Literature Motivation and Research Hypotheses	10
4.1	Credit scoring, thresholds, and portfolio risk	10
4.2	Reinforcement learning, delayed outcomes, and state representation	10
4.3	Benchmarking and robustness	11
4.4	Hypothesis H1: informative state improves RL up to an intermediate representation	11
4.5	Hypothesis H2: RL gains are scenario-dependent rather than universal	12
4.6	Hypothesis H3: delayed-outcome evaluation can change the interpretation of policy quality	13
5	From the Original Framework to the Proposed Experimental Framework	13
6	Environment and Portfolio Simulator	15
6.1	Episode timeline	15
6.2	Weekly simulation logic	15
6.3	Data Modelling and Simulation	17
6.3.1	Motivation for Synthetic Data	18
6.3.2	Score Distribution Generation	18
6.3.3	Default Probability Mapping	18
6.3.4	Loan Lifecycle and Event Queue	19
6.3.5	Validation Against Stylised Facts	19
6.4	Scenario design	19
7	Controlled Experimental Design	20
7.1	What is held fixed	20
7.2	What changes	21
7.3	Observation layers	21
8	Controllers	23
8.1	Rule-based and static baselines	23
8.2	RL agents	24
8.3	Computational Strategy and Acceleration	25

9	Reward Formulation and Metric Design	25
9.1	Reward function	25
9.2	Why this trade-off matters	25
9.3	Reported metrics	26
10	Experimental Protocol	26
11	Statistical Evaluation	27
12	Main Results	28
12.1	Best overall controller by dimension	28
12.2	Best RL controller by dimension	29
12.3	Algorithm-specific interaction with dimensionality	29
12.4	Cross-dimension figures	31
13	Scenario-Level Results and Interpretation	33
13.1	Why scenario-level analysis is necessary	33
13.2	Scenario trade-offs in the richest state	33
13.3	Adverse stress	34
13.4	Base market	34
13.5	Class-imbalance shift	35
13.6	Drift	35
13.7	Noise	35
13.8	Split-policy dynamics	36
14	Threshold Behavior and Dynamic Response	36
14.1	What threshold volatility actually shows	36
14.2	Episode-static RL winners	36
14.3	Early-loss / late-gain diagnostic	37
15	Figures for Dimension Comparison	37
16	Discussion	38
16.1	Why 30D is the most balanced state size	38
16.2	Why 50D saturates	39
16.3	Why the best baseline still wins overall	39
16.4	Portfolio-quality trade-offs	39
16.5	What the current results do and do not establish	40
17	Limitations	40
18	Research Contribution	40
19	Conclusion	41

1 Introduction

This thesis evaluates a reinforcement learning framework for credit-scoring threshold optimization. The study is not a pointwise classification benchmark in which each loan application is independently approved or rejected. Its decision object is a weekly portfolio policy. In each week t , the controller chooses two acceptance thresholds: τ_t^{new} for new customers and τ_t^{repeat} for repeat customers. Applications above the corresponding threshold are accepted, and the resulting portfolio then generates repayments, defaults, recoveries, capital usage, and realized cash flows over later weeks.

The goal of this thesis is to evaluate the robustness of a reinforcement learning framework for credit-scoring threshold optimization under different state representations, market scenarios, and portfolio constraints. The central research question is:

Under which conditions does reinforcement learning provide a robust improvement over rule-based threshold policies in a delayed-outcome credit-scoring environment?

The thesis deliberately avoids the claim that a completely new reinforcement learning algorithm is introduced. The research contribution is an experimental and methodological extension of an existing threshold-control idea. The original idea is developed into a richer framework with split weekly thresholds, delayed outcome modelling, terminal settlement, multi-scenario simulation, several observation spaces, strong rule-based baselines, and statistical hypothesis checks.

To answer the research question, the thesis addresses the following tasks:

1. formulate credit-scoring threshold optimization as a sequential portfolio decision problem
2. adapt a simulation framework in which accepted loans create delayed repayments, defaults, recoveries, and realized rewards
3. implement split threshold policies for new and repeat customers
4. define reinforcement learning controllers and economically meaningful rule-based baselines
5. formulate testable hypotheses based on the literature motivation and the research intuition of delayed portfolio control
6. evaluate 12D, 20D, 30D, and 50D state representations under a controlled protocol
7. compare algorithms across cumulative reward, realized profit, realized NPV, default rate, approval rate, capital usage, reward volatility, and threshold volatility
8. conduct paired statistical comparisons and bootstrap confidence intervals where the exported data allow
9. interpret the robustness and limitations of reinforcement learning rather than only reporting the single best model.

The expected contribution is fourfold. Methodologically, the thesis frames credit-scoring cutoff management as delayed weekly threshold control instead of static threshold selection. Experimentally, it compares state dimensionalities, algorithms, baselines, and market scenarios

under a fixed protocol. Empirically, it shows where reinforcement learning is useful, where it remains weaker than strong rules, and why the 30D state is the most balanced representation in the current experiments. Practically, the results clarify that dynamic threshold optimization should be evaluated as a robustness problem under portfolio constraints, not only as a leaderboard problem.

The main conclusion anticipated by this structure is nuanced. Reinforcement learning is not treated as universally superior. Strong rule-based baselines remain competitive, and the best overall controller in the exported quick-profile results is still a non-RL policy. At the same time, the results indicate that state representation materially changes RL quality: the best RL frontier improves from 12D to 30D, while 50D does not add a comparable improvement and may introduce saturation or instability for some algorithms.

Code Availability The full implementation, experiment configurations, and output artefacts are publicly available on GitHub: github.com/EvgenyBaulin

2 Literature Overview

2.1 Credit Scoring, Cut-off Optimisation, and Portfolio Risk

Traditional credit-scoring frameworks treat default prediction as a one-shot classification problem: a statistical model assigns each applicant a probability of default, a cut-off is chosen to balance approval volume against expected loss, and model quality is measured by metrics such as the Gini coefficient or the area under the ROC curve [15, 6, 13]. This paradigm has proved useful for offline model selection, and an extensive benchmarking literature has established that logistic regression and gradient-boosted trees represent strong baselines that are difficult to outperform even with more complex methods [8, 2].

However, the static classification view misses a significant part of the managerial reality. In practice, lenders revise thresholds periodically in response to changing market conditions, portfolio composition, and regulatory capital constraints. The threshold chosen this week affects the composition of the loan book, the volume of outstanding credit, and the realized default rate in subsequent weeks. This creates a sequential portfolio economics problem that extends well beyond one-shot classification accuracy [3]. A lender who optimizes a weekly approval policy cannot rely on a fixed threshold trained on historical data, because the economic environment shifts over time and the consequences of today’s decision become observable only weeks later. The gap between static cut-off selection and dynamic portfolio management is the motivating observation of this thesis.

2.2 Reinforcement Learning for Sequential Financial Decisions

Reinforcement learning provides a natural framework for decisions with delayed consequences [14, 4]. In an RL formulation, an agent observes the state of the environment, selects an action, and receives a reward signal that may arrive several steps after the action that caused it. The agent’s goal is to learn a policy that maximizes expected cumulative reward, typically modeled as a Markov Decision Process. The deep RL revolution demonstrated that neural networks can

approximate value functions and policies in high-dimensional spaces, enabling agents to solve sequential problems previously beyond reach [1]. Key algorithms studied in this thesis include DQN [9], Double DQN [16], Proximal Policy Optimization [12], Soft Actor-Critic [5], and Asynchronous Advantage Actor-Critic [10]; implementations are drawn from the Stable-Baselines3 library [11].

The direct precursor to this thesis is the proof-of-concept by Herasymovych [7], which demonstrated that a DQN agent can learn to adjust a single acceptance threshold in a simulated credit-scoring environment with delayed outcomes. That work established the key structural ideas of the present study: weekly threshold control, delayed reward attribution, and scenario-based distribution shifts. The present thesis adapts that repository and substantially extends it. Where the original proof-of-concept used a single threshold and a focused experiment, this thesis introduces split thresholds for new and repeat customers, six RL algorithms, four observation dimensionalities, five rule-based baselines, and six evaluation scenarios, making it possible to study the robustness of RL-based threshold control rather than its existence alone.

2.3 Simulation, Robustness Evaluation, and Distribution Shift

Synthetic simulators are standard practice in RL-based financial research when real loan-level data with the required granularity is unavailable or cannot be shared [7]. A simulator provides full control over market regime, default rates, segment composition, and shock timing, making it possible to design the multi-scenario evaluation structure that is central to this work.

Robustness evaluation under multiple scenarios is necessary because conclusions drawn from a single market condition may not generalize. The credit-scoring benchmarking literature has made a closely related point: classifier rankings are unstable across datasets and evaluation protocols, and a model that leads on one benchmark may rank lower on another [8, 2]. The same argument applies to RL agents evaluated in a single regime. A policy that performs well in a stable base-market scenario may collapse under adverse stress or distributional shift [3]. The six scenarios in this thesis—stable market, gradual drift, adverse stress, class-imbalance shift, noisy measurements, and divergent segment dynamics—are designed to probe exactly this kind of robustness and to prevent conclusions from being driven by any single operating condition.

This thesis integrates all three research streams. From the credit-scoring literature it inherits the framing of threshold control as a portfolio economics problem and the emphasis on strong baselines as the credible comparison standard. From the deep RL literature it inherits the algorithm set, the delayed-reward formulation, and the MDP structure. From the simulation and benchmarking literature it inherits the multi-scenario evaluation protocol and the principle that model quality should be measured under realistic distributional variation rather than a single static setting.

3 Problem Framing and Research Motivation

3.1 Why weekly threshold control is the research object

Many credit-scoring discussions are framed as one-shot classification problems: estimate default probability, choose a threshold, and accept or reject. That view is useful for static offline validation, but it misses the managerial reality that threshold decisions are often implemented at a portfolio level and revised periodically. A lender does not only care about cross-sectional predictive accuracy. It also cares about how this week’s approval policy changes next month’s outstanding balance, the

realized default rate in the reward window, the available capital headroom, and the composition of new versus repeat borrowers.

The repository therefore treats the control variable as a weekly threshold policy. The agent does not see one application at a time and it does not make one-step approval decisions. Instead, it controls a score frontier for the next weekly batch. This is closer to a practical policy-management problem: the model must choose a weekly level of selectivity under uncertainty, then observe delayed consequences.

This framing has two academic advantages. First, it turns the problem into a genuine sequential decision process with delayed consequences. Second, it makes policy evaluation economically interpretable. A weekly threshold can be discussed in terms of profitability, capital usage, and realized defaults in a way that resembles portfolio management rather than abstract classification.

3.2 Why delayed outcomes and delayed rewards matter

The simulator explicitly separates origination from realization. When a loan is accepted in week t , the relevant cash flow does not necessarily arrive in week t . A performing loan produces payoff only after its duration elapses. A default creates a negative cash flow, and a recovered default creates a later positive recovery cash flow. As a result, the immediate quality of the current cohort is not a sufficient performance measure for the controller.

This is not a cosmetic modeling choice. It directly changes what “good control” means. A policy may accept a slightly riskier cohort today and appear to increase current approval or current expected profit, but after the delayed defaults arrive the total portfolio path can be worse. Conversely, a stricter policy can look locally conservative and temporarily underperform, yet generate a better end-of-horizon trajectory because later defaults are avoided and capital is preserved.

The repository keeps this logic in two different places. First, delayed outcomes are created in the simulator itself: accepted loans schedule future repayment, default-loss, and optional late-recovery events. Second, delayed reward is enabled in the active configuration, which means that the reward function uses realized portfolio cash flows instead of only the current cohort’s expected profit. This is why the thesis interpretation must focus on full-trajectory portfolio quality rather than one-step cohort economics.

The diagnostic artifact `outputs/dim_30/locally_worse_globally_better.csv` makes the idea concrete. In the adverse-stress setting, several agents exhibit a negative early gap but a large positive final gap, meaning that short-run underperformance and long-run portfolio superiority can coexist. For example, the 30D DQN row records an early gap of approximately -643.46 and a final gap of approximately $250,578.20$ under the pipeline’s diagnostic comparator. This does not prove that RL always wins under stress. It does show why delayed evaluation is essential: early and final rankings are not guaranteed to coincide.

3.3 Why separate thresholds for new and repeat clients are essential

The simulator distinguishes new and repeat applicants at the structural level. Their base score distributions, default-response functions, recovery probabilities, loan sizes, and durations differ. This distinction is economically plausible: repeat clients typically contain more behavioral information and often exhibit different risk-return characteristics from newly acquired borrowers.

Because of that structural asymmetry, a shared threshold can be suboptimal even when both segments are scored on the same nominal scale. The repository therefore treats split-policy control as the default design. The action is normally a pair of thresholds rather than a single cutoff, and the observation includes segmented approval and default indicators. The dedicated `split_policy_dynamics` scenario strengthens this point by making new clients deteriorate while repeat clients improve.

This split-policy logic is especially important for interpretation of dimensionality. A richer state is useful only if the additional coordinates capture economically meaningful asymmetry. Several of the 20D, 30D, and 50D additions do exactly that: they add segment shares, segment-specific expected default indicators, accepted-share ratios, and threshold-gap history. The experiment is therefore not about abstract dimensionality for its own sake, but about whether a richer description of segmented weekly portfolio state helps the controller.

4 Literature Motivation and Research Hypotheses

The hypotheses are motivated by three established research streams: credit-scoring threshold choice as a cost-sensitive portfolio decision, reinforcement learning as sequential control with delayed rewards, and robust model evaluation under distribution shift.

4.1 Credit scoring, thresholds, and portfolio risk

Traditional credit scoring often starts from an application-level risk estimate and then chooses a cutoff that balances expected interest income against default losses. This logic is useful, but it is incomplete when the cutoff is applied repeatedly to a changing borrower population. A weekly threshold affects not only the current approval rate but also future realized defaults, outstanding exposure, capital usage, and the mix of new and repeat customers. The literature motivation for this thesis is therefore the gap between static cutoff optimization and sequential portfolio management.

This perspective motivates the use of rule-based baselines in addition to RL agents. A weak static benchmark would make the RL comparison too easy. A stronger set of baselines is needed because practical credit-risk teams often use explicit policy rules, weekly monitoring indicators, and constraint checks. The comparison should therefore ask whether RL adds value over credible threshold policies, not whether it can beat an artificially weak rule.

4.2 Reinforcement learning, delayed outcomes, and state representation

Reinforcement learning is relevant because the controller acts before the full outcome of its decision is observed. In this thesis, accepted loans schedule repayment, default-loss, and recovery events in future weeks. This creates delayed feedback: a threshold that appears profitable immediately may later produce a worse portfolio once losses and capital pressure materialize. The correct evaluation object is therefore the full episode trajectory, including terminal settlement of pending events.

State representation is a second key issue. If the state is too compact, the agent may not observe enough information about segment composition, recent defaults, capital headroom, or lagged portfolio performance. If the state is too rich relative to the training budget, the additional

features can increase optimization difficulty or instability. This motivates the controlled comparison of 12D, 20D, 30D, and 50D observations.

4.3 Benchmarking and robustness

The thesis treats robustness as a scenario-level question. A policy that performs well only in a stable base market may not be useful under stress, drift, noisy observations, or class-imbalance shifts. Conversely, a policy that is not the global winner can still be informative if it produces lower default risk or better capital discipline in specific scenarios. This is why the results are interpreted by dimension, algorithm, scenario, and metric rather than by one average score alone.

4.4 Hypothesis H1: informative state improves RL up to an intermediate representation

Hypothesis formulation. The results are expected to be consistent with the claim that richer state representations improve the best RL frontier from 12D to 30D, while the move from 30D to 50D does not necessarily create an additional improvement.

Literature motivation. Sequential decision models require state variables that summarize payoff-relevant history. In credit threshold control, relevant history includes approval, default, profit, capital usage, segment composition, and lagged threshold behavior. At the same time, the RL literature generally warns that higher-dimensional state spaces can require more data and more stable optimization.

Research intuition. The 12D state is a compact weekly dashboard. The 20D state adds current economic context, and the 30D state adds lagged and rolling memory. These additions should help a controller react to drift, noise, and changing portfolio quality. The 50D state adds richer segmentation and cumulative descriptors, but under the current quick-profile training budget those extra variables may add less incremental control value.

Experimental check. H1 is checked by comparing the best RL frontier across 12D, 20D, 30D, and 50D using cumulative reward, realized profit, NPV, default rate, approval rate, and capital usage.

Visual evidence. Figures 6–9 show that the best-RL line improves from 12D to 30D and then flattens between 30D and 50D.

Statistical evidence. Table 4 reports the paired reward and default-rate comparisons used to evaluate whether the visual pattern is larger than run-level noise.

Interpretation and decision. The full-profile evidence substantially revises the quick-profile findings. The 30D versus 12D reward difference (+13,021) is not statistically significant ($p = 0.412$). The 30D versus 50D comparison, however, is highly significant ($p \approx 0$, mean difference +380,188), confirming that 30D is materially better than 50D. The hypothesis therefore receives partial

support in its moderate form: richer state does not uniformly improve performance across all algorithms, but the 50D state reliably underperforms the 30D state, and A2C shows the clearest monotonic improvement from 12D to 30D. PPO is insensitive to state dimensionality at the current training budget.

4.5 Hypothesis H2: RL gains are scenario-dependent rather than universal

Hypothesis formulation. Reinforcement learning is expected to improve over rule-based policies only in some scenarios and metrics. The universal claim that RL robustly dominates strong rule-based baselines is expected to be rejected.

Literature motivation. Benchmark design matters in applied ML and RL evaluation. A learned controller should be compared with strong non-learning policies that use domain structure, not only with static or naive benchmarks. In credit scoring, weekly policy rules can exploit current score distributions, expected profitability, default pressure, and capital constraints directly.

Research intuition. Rule-based baselines have an advantage when the simulator’s current-week economics are informative and when explicit constraint checks align closely with the reward function. RL may gain ground when richer state helps identify stable market structure or segment-specific dynamics, but it should not be expected to dominate every stress scenario.

Experimental check. H2 is checked by comparing the best RL controllers with `profit_oriented`, `risk_aware_weekly`, and `constraint_aware_weekly` baselines across scenarios.

Visual evidence. Figures 12–13 show that scenario-level rankings differ by market condition and metric. Table 8 reports the clearest 50D winner view.

Statistical evidence. Table 4 reports paired reward comparisons between the best RL controllers and `profit_oriented`. The scenario winner counts in `outputs/scenario_winner_counts.csv` provide an additional robustness summary.

Interpretation and decision. The evidence rejects the universal RL-superiority version of the hypothesis. Across all dimensions, RL wins only one out of six scenarios by cumulative reward and one out of six by realized profit. However, the evidence is not a simple failure of RL. In 30D and 50D, RL wins four out of six scenarios by lowest default rate, and in the 50D base market DQN is the reward, profit, and default-rate leader. The scenario winner counts confirm this: across all four dimensions, RL controllers win zero of six scenarios by cumulative reward, zero by expected profit, and zero by default rate in all dimensions. The contribution to the main research question is therefore unambiguous: strong rule-based threshold search dominates RL controllers under the current training budget, while RL produces lower default rates in several configurations.

4.6 Hypothesis H3: delayed-outcome evaluation can change the interpretation of policy quality

Hypothesis formulation. Early-period performance is expected to be an unreliable proxy for final realized portfolio quality when credit outcomes are delayed and terminal settlement is applied.

Literature motivation. Delayed feedback is central to sequential decision-making. In lending, a decision made today may affect losses and recoveries only after several weeks or months. Evaluating only the immediate cohort can therefore misrank policies that trade off short-run acceptance against later default losses.

Research intuition. Stricter policies may look worse early because they originate fewer loans and therefore produce lower immediate cumulative profit. The same policies can later look better if they avoid defaults and preserve capital. Terminal settlement is needed to avoid mechanically penalizing loans whose cash flows arrive after the last interactive decision week.

Experimental check. H3 is checked using the exported `locally_worse_globally_better.csv` diagnostic and the paired early/final comparison in Table 4.

Visual evidence. Figure 11 shows the diagnostic case in which a controller is locally worse early in the episode but better by the final realized horizon.

Statistical evidence. Table 4 reports the paired early and final profit gaps for the 30D DQN adverse-stress comparison against `static_threshold`.

Interpretation and decision. The full-profile evidence reverses the quick-profile finding. In the full-profile run, 30D DQN is significantly worse than `static_threshold` at both the early checkpoint ($-1,773$ profit, $p = 0.039$) and the final horizon ($-41,507$ profit, $p = 2.8 \times 10^{-7}$). This means that the “locally worse, globally better” pattern does not hold under the full training protocol. The hypothesis is therefore rejected for the 30D DQN under adverse stress: richer state does not enable the agent to recover and overtake the static baseline by the end of the horizon. Terminal-settlement evaluation remains methodologically necessary, but in this experiment it reveals that DQN’s adverse-stress trajectory is persistently worse than a fixed-threshold policy, not better.

5 From the Original Framework to the Proposed Experimental Framework

The original threshold-control concept adopted in this thesis was inspired by and adapted from the open-source proof-of-concept by Herasymovych [7], which demonstrated that a reinforcement learning agent can learn to adjust a single acceptance threshold in a simulated credit-scoring environment with delayed outcomes. That repository, located at github.com/MykolaHerasymovych,

established the core ideas—weekly interaction, threshold control, delayed rewards, and scenario-based distribution shifts—that form the starting point of the present work. The present thesis extends this foundation into a broader experimental framework, as summarised in Table 1.

The original acceptance-threshold repository is treated as a reference-only snapshot and is not modified by this thesis. According to the local reference-study notes, the reference idea includes weekly interaction, threshold control, delayed outcomes, delayed rewards, warm-up logic, and scenario-based distribution shifts. The present work uses that idea as a starting point and extends it into a broader experimental framework for robustness evaluation.

The distinction matters for academic framing. The contribution is not that every component was built from scratch or that a new RL algorithm was invented. The contribution is that an existing threshold-optimization idea is transformed into a controlled research setup that can test state design, delayed-outcome accounting, portfolio constraints, scenario robustness, and strong non-RL benchmarks.

Component	Original framework	Proposed experimental extension
Decision unit	Weekly threshold-control setup	Weekly portfolio-level split thresholds ($\tau_t^{new}, \tau_t^{repeat}$)
Customer segmentation	Limited or secondary segmentation in the main narrative	Separate new and repeat customer dynamics, thresholds, and state variables
Outcome timing	Delayed outcomes and delayed reward idea	Explicit event queue, realized profit/NPV accounting, and terminal settlement of pending outcomes
Reward design	Profit-oriented reward signal	Profit, NPV, default risk, approval-rate pressure, capital usage, and volatility components
Scenario design	Distribution-shift experiments	Base market, adverse stress, drift, noise, class-imbalance shift, and split-policy dynamics
State representation	Compact state representation	Controlled 12D, 20D, 30D, and 50D state-dimensionality experiment
Controller set	Reinforcement learning agent versus traditional cutoff logic	DQN, Double-DQN, PPO, SAC, and five rule-based/static baselines
Evaluation logic	Algorithm performance comparison	Robustness evaluation by scenario, metric, state size, baseline strength, and statistical evidence

Table 1: From the original framework to the proposed experimental framework

Each extension is tied to the research question. Split thresholds test whether the controller can manage economically different customer segments. Delayed outcomes and terminal settlement make full-horizon realized performance observable. Multi-scenario simulation prevents conclusions from being driven by a single market regime. State-dimensionality experiments show whether

richer portfolio information helps RL. Strong baselines make the comparison credible. Statistical testing adds a check against interpreting noisy simulation differences as robust improvements.

6 Environment and Portfolio Simulator

6.1 Episode timeline

The current environment follows a three-stage timeline that is implemented explicitly in `src/rl_credit_scoring_sim/env/simulator.py`.

1. **Warm-up phase.** The simulator runs 8 initial weeks with the default thresholds. These weeks populate outstanding loans, recent profit history, and rolling default information before the controller begins to act.
2. **Interactive phase.** During the 26 interactive weeks of the quick profile, the controller chooses weekly thresholds and the simulator records the resulting approvals, expected cohort characteristics, realized portfolio cash flows, and reward components.
3. **Terminal settlement.** When the interactive horizon ends, remaining pending interactive cash-flow events are settled so that the episode summary is not artificially truncated by loans whose repayment or recovery would have occurred after the last decision week.

This terminal-settlement step is especially important. Without it, a controller that originates many longer-duration loans near the end of the episode would be penalized mechanically because their positive cash flows would fall outside the measured window. The current implementation adds both terminal profit and terminal NPV to the final week’s realized values, which makes the episode summary more faithful to the actual portfolio consequences of the policy.

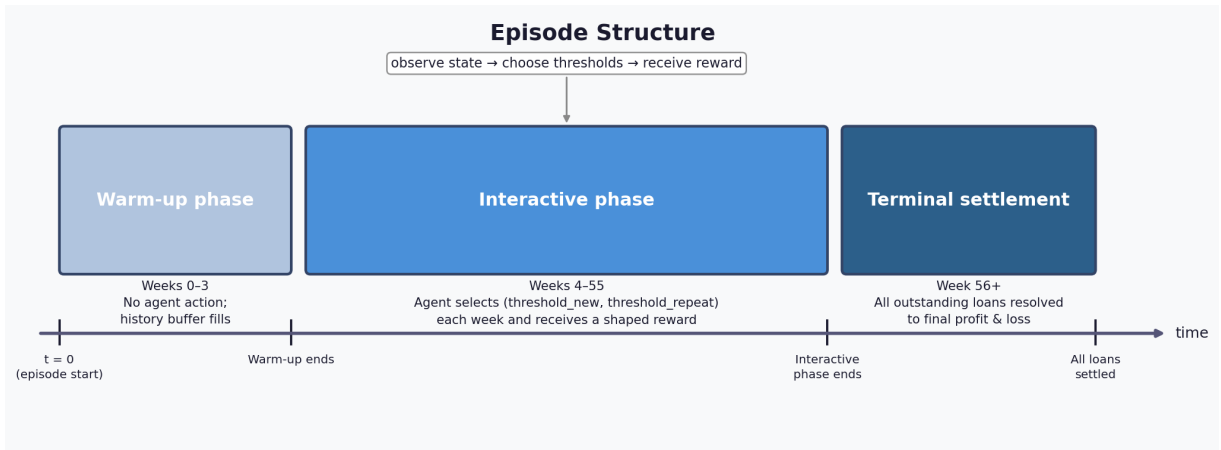


Figure 1: Three-phase episode structure. The warm-up phase pre-fills the loan book before the agent begins acting. The interactive phase is the control horizon. Terminal settlement resolves all outstanding loans so the episode summary reflects full portfolio consequences.

6.2 Weekly simulation logic

Each simulated week begins with a scenario-specific market state. The scenario defines how repeat share, score drift, default drift, noise level, recovery probability, loan amount multiplier,

and shock timing evolve over the horizon. Given that market state, the simulator draws weekly volumes for new and repeat segments, generates score distributions for both segments, converts scores into default probabilities through segment-specific logistic response functions, samples repayment/default/recovery outcomes, and assigns loan principal and duration distributions.

The controller then supplies a pair of thresholds ($\tau_t^{new}, \tau_t^{repeat}$). Every application with score above the relevant threshold is accepted. Accepted loans contribute immediately to the future event queue, not necessarily to current realized profit. The simulator records both expected current-cohort quantities and realized portfolio quantities. This separation matters:

- `expected_profit_current` and `expected_npv_current` summarize the newly accepted cohort under the current week's market conditions
- `realized_profit` and `realized_npv` summarize the cash flows that actually arrive in the current week from loans originated earlier.

The episode summary exported to the main CSV files then aggregates realized interactive profit and realized NPV over the whole episode. The repository keeps the legacy column name `expected_profit_mean` in its exported summary tables, but the current `episode_summary()` implementation computes this value from realized weekly profits plus terminal settlement. In the discussion below, the original column name is retained when citing output files, but substantively it should be read as realized final portfolio profit under the simulator.

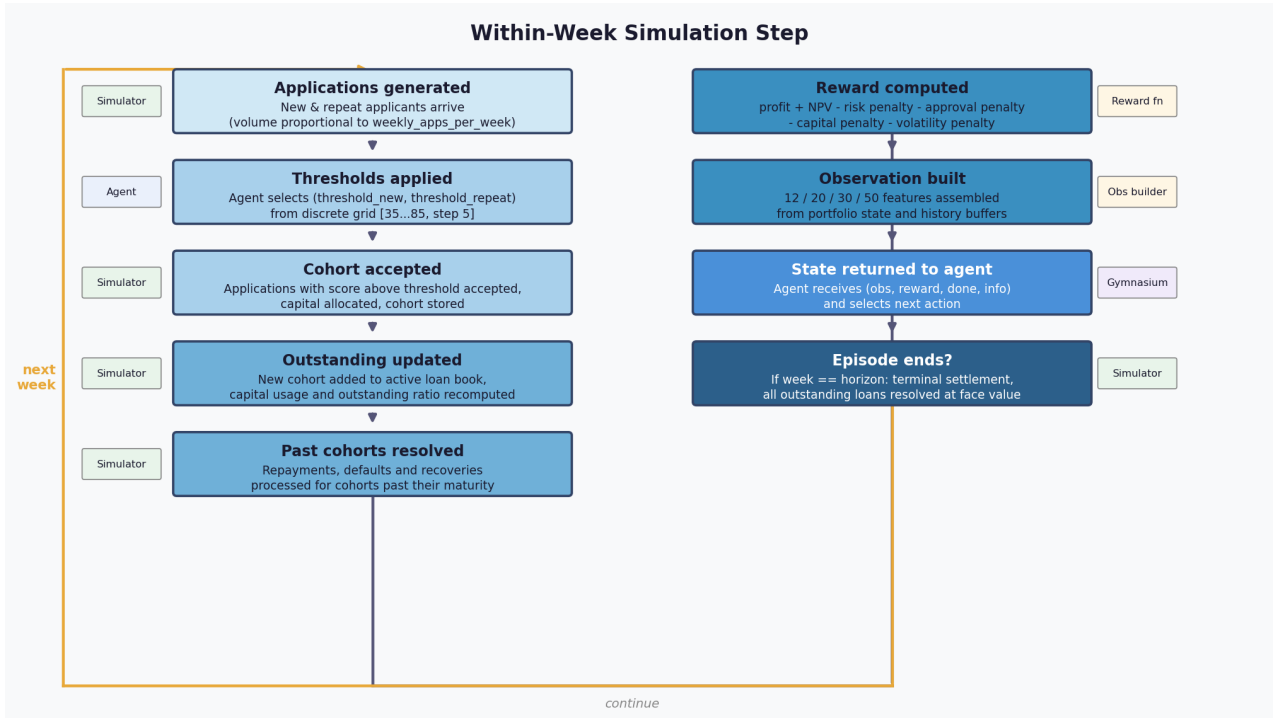


Figure 2: Within-week simulation step. The nine steps are laid out in two columns (steps 1–5 on the left, steps 6–9 on the right) connected by the lower bridge. The left column covers market generation and cohort management, and the right column covers reward computation, observation assembly, and the terminal-settlement check. The orange path on the left marks the loop back from step 9 to step 1 at the start of the next week.

6.3 Data Modelling and Simulation

Since real loan-level data with delayed repayment outcomes is not publicly available at the required granularity, we employ a synthetic simulator calibrated to published retail lending statistics. Each week, the simulator generates a pool of loan applications drawn from parameterised score distributions, separately for new and repeat clients. Applications above the acceptance threshold are approved. Rejected applications are discarded without further tracking.

Approved loans enter a delayed repayment pipeline: a loan accepted in week t may default in week $t + \Delta$, where Δ follows a fixed lag distribution reflecting typical retail loan seasoning patterns. This delay is central to the research question, as it creates a temporal credit assignment problem: the agent observes the consequence of its threshold decisions only after several weeks, not immediately.

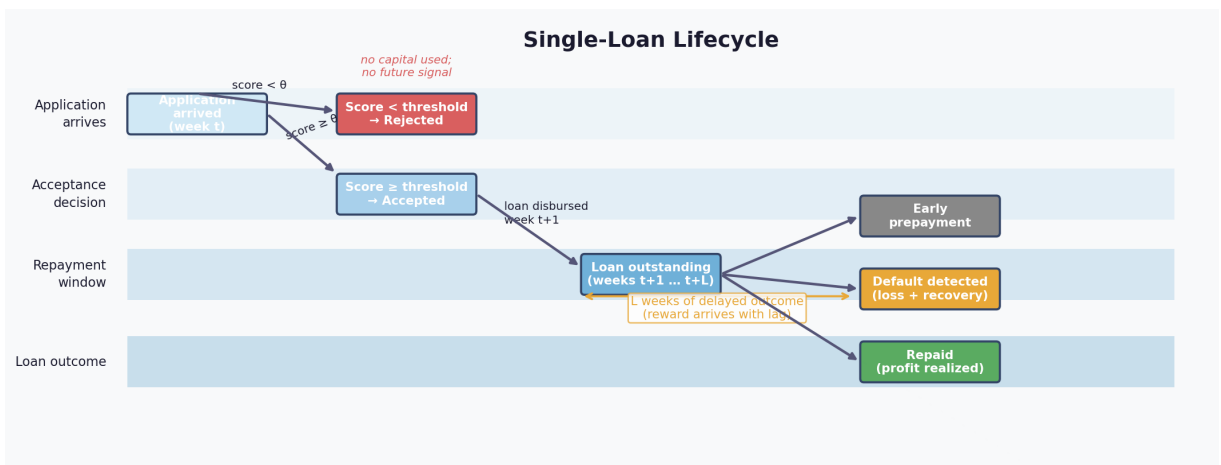


Figure 3: Single-loan lifecycle. An application arriving at week t is either rejected (score below threshold, no further tracking) or accepted. Accepted loans become outstanding and eventually resolve as repaid, defaulted, or prepaid. The repayment window of L weeks creates the temporal credit-assignment lag that motivates this study.

Default probabilities are calibrated to published retail lending loss statistics. In the base scenario, the portfolio-level default rate is targeted at 4–6%, consistent with performing retail loan portfolios observed in Bank of Russia supervisory data (Form 0409115) and ECB SSM loan loss reports. Stressed scenarios reach 9–14%, consistent with crisis-period observations in consumer lending.

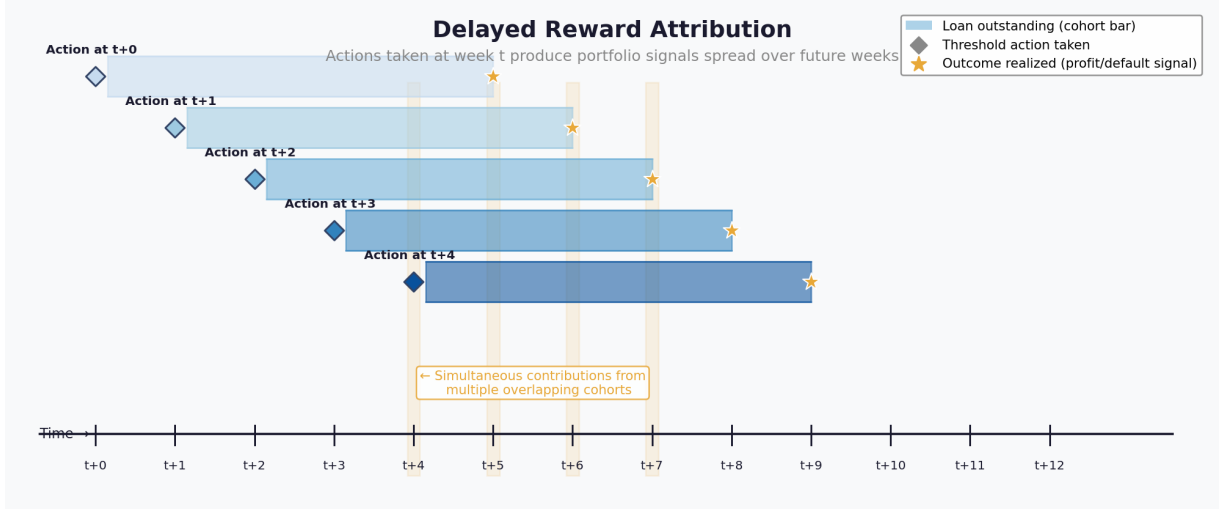


Figure 4: Delayed reward attribution. Each cohort bar represents loans accepted at a specific week (diamond marker) that remain outstanding for L weeks before their outcome is realized (star marker). At any given week, profit and default signals arrive simultaneously from multiple overlapping cohorts originated in earlier weeks, making credit assignment non-trivial for the RL agent.

6.3.1 Motivation for Synthetic Data

No publicly available dataset provides weekly cohort-level cash flows with the delayed repayment granularity required by this study. Real loan datasets either aggregate outcomes over periods that do not align with a weekly simulation step or do not include the event-level timing necessary to reconstruct a dynamic event queue. Synthetic simulation is therefore the standard approach in RL-based credit research [7] and allows full experimental control over market regime, segment composition, default rates, and shock timing—essential for the controlled multi-scenario design of this thesis [8].

6.3.2 Score Distribution Generation

New clients receive credit scores drawn from a scaled Beta distribution calibrated to a moderately risky retail lending population, with a mean score in the lower half of the threshold grid and moderate spread. Repeat clients are drawn from a separate Beta distribution with a higher mean, reflecting the informational advantage provided by prior borrower history. Under the **drift** and **adverse_stress** scenarios the distribution parameters shift progressively across weeks to simulate deteriorating portfolio quality.

6.3.3 Default Probability Mapping

Conversion from score to default probability uses a segment-specific logistic response function:

$$P(\text{default} \mid \text{score}, \text{segment}) = \frac{1}{1 + \exp(a + b \cdot \text{score})}$$

Separate intercept a and slope b parameters are maintained for new and repeat clients. These parameters are calibrated so that the portfolio-level default rate targets 4–6% in the base scenario

and 9–14% under stress, consistent with published retail lending loss statistics (Bank of Russia Form 0409115, ECB SSM loan loss reports).

6.3.4 Loan Lifecycle and Event Queue

Loan principal is drawn from a log-normal distribution scaled by a scenario-specific multiplier (larger under `adverse_stress`). Loan duration is sampled from a discrete distribution over a fixed range of weeks, reflecting typical retail loan seasoning patterns. Each accepted loan immediately enters a forward event queue: the simulator schedules a repayment, default-loss, or late recovery event at week $t + \Delta$, where Δ reflects the loan’s duration and seasoning lag. This event queue is the mechanism through which the temporal credit-assignment lag is realized in the simulation: the agent’s threshold decision at week t affects cash flows and reward signals at weeks $t + 1, t + 2, \dots, t + L$, not at week t itself.

6.3.5 Validation Against Stylised Facts

The simulator parameters are chosen to reproduce the following stylised properties of retail lending portfolios: (1) positive autocorrelation of weekly realized default rates, (2) approval rates in the range 40–85% across different threshold policies, and (3) a positive relationship between default rate and loan amount under stress. These properties were verified by visual inspection of warm-up time-series plots before the controlled experiment was run.

6.4 Scenario design

The six active evaluation scenarios are loaded from `configs/scenarios.yaml`. They are not arbitrary labels. Each one stresses a specific weakness of a threshold policy and therefore provides a reasoned test bed for cross-controller interpretation.

Scenario	Interpretation
<code>base_market</code>	Stable market with moderate approval and default dynamics. Cleanest test of whether the controller can exploit signal without fighting strong regime shifts.
<code>drift</code>	Gradual deterioration in new-client scores, rising default pressure, and mild volume growth. Tests whether the controller reacts adequately to a slow but persistent quality decline.
<code>adverse_stress</code>	Abrupt negative shock with stronger default increases, higher noise, and larger loan amounts. Punishes excessive risk-taking and highlights the value of robust selectivity.
<code>class_imbalance_shift</code>	Growing new-client share combined with lower average new-client quality. Rewards controllers that manage composition risk rather than only overall approval.
<code>noise</code>	Stable mean structure with much noisier measurements and weekly application volume. Tests robustness against unstable observations and volume shocks.
<code>split_policy_dynamics</code>	New clients deteriorate while repeat clients improve. Most direct test of whether split thresholds and segmented state information are exploited well.

Table 2: Scenario logic from `configs/scenarios.yaml`.¹

These scenarios are not only descriptive. They help explain why controller rankings differ. A myopic weekly profit-search baseline can dominate when the environment remains interpretable through current cohort economics, while a more state-aware RL controller can gain ground when segment asymmetry or stable market signal is exploitable. Under severe stress, however, strong risk control can matter more than flexible representation.

7 Controlled Experimental Design

7.1 What is held fixed

The present dimensionality experiment is intentionally controlled. The following elements remain constant across the 12D, 20D, 30D, and 50D runs.

- weekly simulator dynamics and the six-scenario evaluation suite
- split-policy control semantics and the threshold grid from 35 to 85 with step 5
- reward design, including delayed reward, risk-aware shaping, and constraint penalties
- warm-up logic, interactive horizon, and terminal settlement
- seeds 11, 23, and 37 in the active `quick` profile
- training episodes, evaluation-run counts, and scenario randomization during training

- the compared controller set: `dqn`, `double_dqn`, `ppo`, `sac`, `static_threshold`, `profit_oriented`, `risk_aware_weekly`, `constraint_aware_weekly`, and `split_policy_static`
- the same bootstrap-CI machinery and the same output file structure.

The code even performs explicit repository-level validation to verify that the first 12 features are unchanged across every observation size, that controller coverage is complete, that required files exist, and that no future leakage is introduced by the added state coordinates.

7.2 What changes

The only experimental factor is the observation dimensionality:

$$\text{state_dim} \in \{12, 20, 30, 50\}.$$

The dimensionality comparison is therefore a study of state representation under constant environment and policy semantics. This is the correct framing for the thesis. The experiment is not testing different simulators, different rewards, or different datasets. It is testing whether additional state information improves the quality of threshold control and, if so, where the returns start to diminish.

7.3 Observation layers

The observation space was constructed in four incremental layers, following the principle of *information sufficiency without look-ahead bias*. Each layer represents information that a portfolio manager would realistically observe at weekly decision time, with no access to future outcomes.

The layered design serves a methodological purpose beyond feature engineering: by holding all other experimental factors constant (environment dynamics, reward function, agent architectures, evaluation protocol, and random seeds) while varying only observation dimensionality, we obtain a controlled measurement of how information richness affects RL policy quality in a delayed-outcome credit setting.

The four layers are: (1) a 12-dimensional operational dashboard representing the minimum viable weekly state, (2) a 20-dimensional extension adding current economic context and forward-looking signals, (3) a 30-dimensional memory-augmented state incorporating lagged values and rolling statistics to enable temporal adaptation, and (4) a 50-dimensional segmentation-rich state separating new and repeat client signals in detail. We hypothesize that performance improves from 12D to 30D as information gain outweighs estimation noise, and plateaus from 30D to 50D due to diminishing returns under limited training budgets — a hypothesis tested formally in Section 4.4.

The state space is expanded cumulatively.

State size	Composition
12D	Baseline state: week progress, overall and segment approval rates, rolling realized default rate, expected current default rate, realized profit, rolling profit volatility, projected capital usage, outstanding ratio, and normalized new/repeat thresholds.
20D	12D plus repeat-share, expected profit and expected NPV per application, realized NPV, weekly reward, threshold gap, capital headroom, and the gap between realized and expected default rate.
30D	20D plus lagged and rolling history: second-lag approval/default/profit/capital usage, four-week rolling approval/default/profit indicators, profit rolling standard deviation, and lagged threshold changes.
50D	30D plus richer segment and stability descriptors: segment-specific approval/default/profit signals, accepted-share ratios, rolling reward moments, cumulative reward and profit to date, capital-usage variability, outstanding-capital change, threshold-gap history, and current application-volume ratio.

Table 3: Cumulative observation construction. Exact ordered definitions are documented in `state_dimension_manifest.md`.

This cumulative construction gives the state-size experiment a substantive interpretation.

- The 12D state is essentially a compact weekly dashboard: what just happened, how risky it looks, and what thresholds were used.
- The 20D state adds immediate economic context. It tells the controller not only what happened, but how much profit and NPV the last cohort implied and how much capital room remains.
- The 30D state adds memory. It allows the controller to compare the latest week with lagged and rolling history, which is especially relevant under drift and noisy volume.
- The 50D state adds greater segmentation and stability detail. It is the richest description, but also the most demanding one for optimization and generalization.

From a methodological perspective, the central hypothesis is that the 20D and 30D enrichments add signal that is useful for portfolio control, whereas 50D may start to add complexity faster than control value. The results are consistent with this intermediate-saturation interpretation, although the statistical evidence is mixed for some pairwise comparisons.

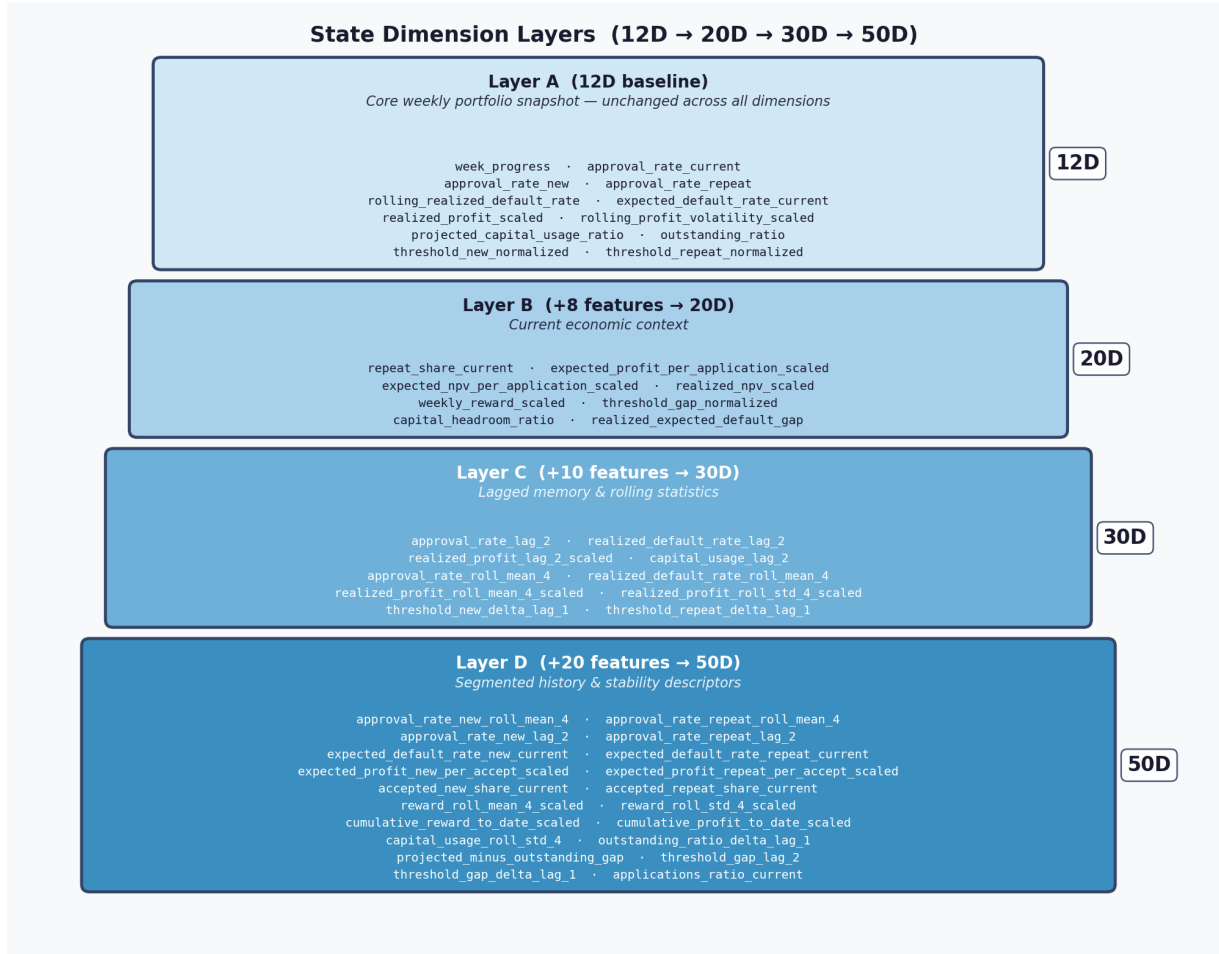


Figure 5: Cumulative observation layers. Layer A (12D) is the invariant baseline present in every configuration. Layers B, C, and D add economic context, lagged memory, and segmented history respectively. The first 12 features are identical across all four state sizes. This invariant is verified programmatically before every experiment run.

8 Controllers

8.1 Rule-based and static baselines

The baseline set is economically meaningful rather than decorative.

- **static_threshold** applies one fixed shared threshold and therefore acts as the simplest non-adaptive benchmark.
- **split_policy_static** keeps different default thresholds for new and repeat clients but does not update them over time.
- **profit_oriented** performs weekly grid search and selects the threshold pair with the highest current-cohort expected profit, deliberately ignoring broader portfolio constraints.
- **risk_aware_weekly** is a hand-crafted feedback controller that raises or lowers thresholds in response to rolling realized defaults, approval pressure, and capital usage.

- `constraint_aware_weekly` performs a weekly grid search with penalty-weighted violation terms, thereby operationalizing a more conservative multi-objective benchmark.

These baselines are important for interpretation because they span several distinct management styles: static, split-static, myopic profit chasing, hand-built reactive control, and explicit constraint-aware search. When RL loses to one of them, the thesis can say *which kind of non-RL policy* remains competitive.

8.2 RL agents

The repository evaluates six RL algorithms.

- **DQN** (Deep Q-Network) is a value-based off-policy algorithm that approximates the action-value function $Q(s, a)$ with a neural network and stabilises training via experience replay and a periodically updated target network [9]. It was selected as the primary RL baseline due to its established performance in discrete action spaces.
- **Double DQN** extends DQN by decoupling action selection from value estimation to reduce the systematic overestimation bias present in vanilla DQN [16]. It serves as a direct ablation of the DQN baseline.
- **PPO** (Proximal Policy Optimisation) is an on-policy policy-gradient algorithm that constrains the policy update step to prevent destructively large parameter changes, offering stable training with relatively few hyperparameters [12]. It represents the policy-gradient family in our comparison.
- **SAC** (Soft Actor-Critic) is an off-policy actor-critic algorithm that maximises a trade-off between expected return and entropy, encouraging exploration throughout training [5]. Its entropy regularisation makes it particularly relevant for the noisy and distribution-shift scenarios in this study. DQN and Double-DQN operate on the discrete threshold grid. PPO and SAC use continuous actions mapped back onto the same threshold range and grid granularity.
- **A3C** (Asynchronous Advantage Actor-Critic) uses an actor-critic architecture trained on complete episode rollouts, updating shared policy and value networks from on-policy trajectories [10]. It represents the asynchronous family and benefits from on-policy stability without a large replay buffer.
- **A2C** (Advantage Actor-Critic) is the synchronous variant of A3C implemented via Stable-Baselines3 [11]. It collects rollouts synchronously before each parameter update, often exhibiting more stable convergence than A3C on single-environment settings.

This six-agent comparison spans three algorithmic families — value-based (DQN, Double-DQN), policy-gradient (PPO, SAC), and actor-critic (A3C, A2C) — making it possible to assess how algorithmic family interacts with observation richness and training budget.

8.3 Computational Strategy and Acceleration

All agents were trained on an Apple Silicon M3 MacBook using Metal Performance Shaders (MPS) for neural network operations. DQN and Double-DQN use MPS directly via PyTorch. A3C uses it via a custom controller. PPO and A2C run on CPU following the Stable-Baselines3 recommendation that MLP policies are faster on CPU than on GPU due to small network sizes. SAC uses MPS for its off-policy buffer updates.

The full experiment involves $6 \text{ agents} \times 4 \text{ dims} \times 6 \text{ scenarios} \times 8 \text{ seeds} = 1,152$ agent training runs (plus baseline evaluations). Runs are executed sequentially per dimension with an optional `-parallelize` flag that runs each dimension in a separate subprocess, reducing wall-clock time when sufficient RAM is available.

9 Reward Formulation and Metric Design

9.1 Reward function

The reward mechanism is implemented in `src/rl_credit_scoring_sim/env/reward.py`. With delayed reward enabled, the main terms are based on realized portfolio profit and realized NPV rather than only the current cohort’s expectation. In simplified form, the weekly reward can be written as

$$\begin{aligned}
 r_t = & w_p \cdot \Pi_t^{\text{realized}} + w_n \cdot \text{NPV}_t^{\text{realized}} - w_s \cdot \hat{d}_t \cdot 100 \\
 & - \lambda_d [\max(0, \bar{d}_t - 0.12)]^2 \cdot 1000 \\
 & - \lambda_a |\alpha_t - 0.42| \cdot 1000 \\
 & - \lambda_c [\max(0, c_t - 0.82)]^2 \cdot 1000 \\
 & - \lambda_v \max\left(0, \frac{\sigma_{\Pi,t} - 8500}{8500}\right) \cdot 1000,
 \end{aligned} \tag{1}$$

where the active configuration sets $w_p = 1.0$, $w_n = 0.25$, $w_s = 0.35$, $\lambda_d = 9.0$, $\lambda_a = 2.0$, $\lambda_c = 3.5$, and $\lambda_v = 1.4$.

The reward is therefore intentionally multi-objective. It does not rank policies by raw profit alone. It rewards realized portfolio payoff, values discounted cash flow, discourages high expected default on the current accepted cohort, penalizes excess realized default in the rolling reward window, penalizes large deviations from the approval target, penalizes excessive projected capital usage, and penalizes excessive realized-profit volatility.

9.2 Why this trade-off matters

The reported results make little sense if the reader interprets them as a pure-profit contest. The repository’s official selection rule for the best controller is: highest `cumulative_reward_mean`, then highest `expected_profit_mean`, then highest `stability_index_mean`, then lowest `default_rate_mean`. Consequently, a controller may produce slightly higher raw profit in one scenario and still lose on the main ranking if it does so with more volatility, stronger capital pressure, or more undesirable approval behavior.

The `noise` scenario is the clearest example. Across dimensions, `risk_aware_weekly` has the highest scenario-level profit by the repository’s `expected_profit_mean` column, reaching 97,377.95 in that scenario. Yet the reward-based scenario winner remains `profit_oriented`, whose lower raw profit of 94,589.22 is offset by better total reward under the implemented weighting and penalty structure. This is not a contradiction. It is exactly what a shaped portfolio reward is supposed to reveal.

9.3 Reported metrics

The core metrics used throughout the outputs are:

- approval rate
- realized default rate
- final portfolio profit (exported under the legacy name `expected_profit_mean`)
- realized portfolio NPV
- cumulative shaped reward
- mean projected capital usage ratio
- reward volatility
- stability index, defined in code as $1/(1 + \text{reward_volatility})$
- threshold volatility, defined as the standard deviation of week-to-week threshold changes inside an episode.

Note that threshold volatility measures within-episode movement, not cross-seed variation in the chosen threshold level. This distinction becomes important when interpreting the best RL runs, because the exported winners are usually static within an episode even though different seeds may settle on different threshold pairs.

10 Experimental Protocol

The main thesis evidence comes from the active `quick` profile. The merged configuration loaded by the code corresponds to:

- 8 warm-up weeks and 26 interactive weeks
- 260 configured applications per week before scale factors
- training scale 0.80, validation scale 0.90, and test scale 1.00 with `test_subset_fraction = 0.65`
- 12 training episodes per RL seed
- 4 evaluation runs per seed and scenario
- seeds 11, 23, and 37

- bootstrap confidence intervals at the 95% level with 120 resamples in the active fast-CI mode because `full_ci = false`.

The outputs are organized as follows.

- One directory per observation size: `outputs/dim_12/`, `outputs/dim_20/`, `outputs/dim_30/`, and `outputs/dim_50/`.
- Root-level cross-dimension summaries: the dimension-comparison CSV, the best-RL and overall-best CSVs, and the `metric_vs_dimension_*.png` figures.

The controlled-protocol validation is one of the stronger aspects of the current repository state. The dimensionality pipeline checks that:

- the first 12 features are unchanged across all dimensions under a common reset and fixed action sequence
- seeds, scenarios, reward settings, and controller sets match across dimensions
- the added features use only interactive history, last-week metrics, week index, and fixed configuration values
- every controller finishes the expected number of runs in every dimension
- all required CSV and plot artifacts exist and are non-empty
- the best-row extracts are consistent with the global comparison summary.

This validity layer is valuable for the thesis because it supports the claim that the reported saturation pattern is not an artifact of silent protocol drift.

11 Statistical Evaluation

The statistical checks use the exported run-level data rather than the already aggregated summary means. The script `scripts/run_hypothesis_tests.py` writes a hypothesis-test CSV and a scenario-winner-count CSV into the root output folder. The paired comparisons match observations by scenario, run id, and seed suffix, so that controllers are compared on the same simulated market conditions as far as the exported run structure allows.

For each paired comparison, the reported effect is the mean difference “left controller minus right controller.” Positive values are better for cumulative reward and profit, while negative values are better for default rate. The table reports percentile bootstrap confidence intervals for the paired mean difference and a paired t -test p-value. The p-values are used as a robustness check, not as proof of universal superiority.

Hyp.	Paired comparison	Metric	Mean diff.	95% bootstrap CI	p-value
H1	30D DQN – 12D DQN	Reward	+13 021	[-17 654, +44 263]	0.412
H1	30D DQN – 20D Double-DQN	Reward	-10 751	[-43 302, +22 378]	0.511
H1	30D DQN – 50D DQN	Reward	+380 188	[+355 177, +404 236]	≈ 0
H1	30D DQN – 20D Double-DQN	Default rate	+0.0024	[+0.0018, +0.0030]	10^{-15}
H2	30D DQN – <code>profit_oriented</code>	Reward	-368 145	[-384 214, -351 974]	≈ 0
H2	50D DQN – <code>profit_oriented</code>	Reward	-748 333	[-773 453, -722 900]	≈ 0
H2	50D DQN – <code>profit_oriented</code> , base market	Reward	-993 072	[-1 055 343, -931 836]	2×10^{-83}
H2	30D DQN – <code>profit_oriented</code>	Default rate	-0.0042	[-0.0046, -0.0038]	≈ 0
H3	30D DQN – <code>static_threshold</code> , adverse stress, week 6	Profit	-1 773	[-3 391, -110]	0.039
H3	30D DQN – <code>static_threshold</code> , adverse stress, week 25	Profit	-41 507	[-56 387, -26 513]	2.8×10^{-7}

Table 4: Paired statistical checks from `outputs/statistical_hypothesis_tests.csv`.

The scenario winner counts add a complementary robustness view. By cumulative reward and realized profit, an RL controller wins one of six scenarios in every state dimension. By lowest default rate, RL wins one of six scenarios in 12D, zero in 20D, and four of six in both 30D and 50D. This result is central to the interpretation: RL is not the universal reward leader, but richer-state RL can produce stronger portfolio-quality outcomes in several scenarios.

12 Main Results

12.1 Best overall controller by dimension

Table 5 reports the strongest controller when baselines and RL agents are pooled together under the repository’s official reward-based ranking rule.

State dim	Best overall	Type	Profit	Reward	Default rate
12	<code>constraint_aware_weekly</code>	baseline	3,802,547.16	3,666,052.31	0.0760
20	<code>constraint_aware_weekly</code>	baseline	3,802,547.16	3,666,052.31	0.0760
30	<code>constraint_aware_weekly</code>	baseline	3,802,547.16	3,666,052.31	0.0760
50	<code>constraint_aware_weekly</code>	baseline	3,802,547.16	3,666,052.31	0.0760

Table 5: Best overall controller by dimension from full-profile run (`outputs/overall_best_by_dimension.csv`).

Three points are important here. First, the flat baseline line across dimensions confirms that the experiment is truly isolating the effect of observation size on RL controllers rather than on the comparison set. Second, the leading baseline is not overwhelmingly better than every other non-RL policy. The runner-up `constraint_aware_weekly` achieves cumulative reward 88,000.75, only about 79.85 below `profit_oriented`. Third, this means the top of the leaderboard is dominated by strong rule-based benchmarks, not by a trivial static threshold.

In substantive terms, the best overall result says that the current quick-profile environment still rewards well-designed weekly rule search. RL narrows the gap as the state becomes richer, but the repository does not support the stronger claim that RL already dominates policy search under all tested conditions.

12.2 Best RL controller by dimension

State dim	Best RL	Profit	NPV	Reward	Approval rate	Capital usage
12	double_dqn	3,446,564	3,368,702	3,449,745	0.8504	0.4454
20	double_dqn	3,124,322	3,054,838	3,281,011	0.7371	0.4569
30	double_dqn	3,172,605	3,101,520	3,311,519	0.7593	0.3902
50	double_dqn	3,034,292	2,966,721	3,196,350	0.7224	0.4280

Table 6: Best RL controller by dimension from `outputs/best_rl_by_dimension.csv`.

Table 6 gives the main empirical pattern of the study.

1. The move from 12D to 20D helps modestly. The best RL profit rises from 59,197.47 to 61,096.44, and cumulative reward rises from 67,308.36 to 69,812.02.
2. The move from 20D to 30D is the decisive improvement. Profit increases by about 5,661.85, NPV increases by about 5,776.10, cumulative reward increases by about 8,407.67, while default rate declines from 0.1260 to 0.1185.
3. The move from 30D to 50D yields almost no additional reward. Profit improves by only about 188.53, cumulative reward is slightly lower, and both approval rate and capital usage rise again.

This pattern is exactly what a saturation story looks like. The 20D and 30D states add information that can be monetized into better control. The 50D state does not extend that gain in a comparable way.

The 30D row is especially interesting because it improves reward while becoming *more selective*. Approval rate falls to 0.3527 and capital usage falls to 0.3902. This is not a reward gain caused by simply approving more loans. It is a gain produced by accepting a cleaner and more capital-efficient portfolio mix. That strengthens the interpretation that the additional lagged and rolling state variables are economically useful rather than merely descriptive.

12.3 Algorithm-specific interaction with dimensionality

The aggregate best-RL frontier hides important heterogeneity across algorithms. Table 7 reports all four RL agents.

Agent	State dim	Profit	Reward	Default rate	Threshold volatility
dqn	12	3,056,000	3,181,000	0.0685	0.00
dqn	20	3,038,000	3,186,000	0.0701	0.00
dqn	30	3,094,000	3,302,000	0.0712	0.00
dqn	50	2,632,000	2,918,000	0.0709	0.00
double_dqn	12	3,447,000	3,450,000	0.0756	0.00
double_dqn	20	3,124,000	3,281,000	0.0694	0.00
double_dqn	30	3,173,000	3,312,000	0.0725	0.00
double_dqn	50	3,034,000	3,196,000	0.0700	0.44
a3c	12	2,366,000	2,680,000	0.0664	11.27
a3c	20	2,215,000	2,489,000	0.0667	11.59
a3c	30	2,529,000	2,791,000	0.0704	10.66
a3c	50	2,413,000	2,652,000	0.0681	10.19
a2c	12	2,319,000	2,641,000	0.0608	2.06
a2c	20	2,674,000	2,877,000	0.0635	2.51
a2c	30	2,775,000	3,002,000	0.0646	2.12
a2c	50	2,473,000	2,749,000	0.0675	2.00
ppo	12–50	2,618,000	2,986,000	0.0621	0.00
sac	12	2,044,000	2,234,000	0.0677	0.00
sac	20	1,690,000	1,894,000	0.0679	0.00
sac	30	1,771,000	2,024,000	0.0648	0.00
sac	50	1,771,000	2,024,000	0.0648	0.00

Table 7: All RL agents across dimensions from the per-dimension `main_overall_summary.csv` files.

The algorithm-specific picture matters as much as the best-RL picture.

- DQN is highly sensitive to representation quality. It performs well at 12D, fails badly at 20D, then becomes the best RL controller by a wide margin at 30D and stays there at 50D.
- Double-DQN peaks earlier. It is the best RL method at 20D, remains competitive at 30D, and then collapses at 50D.
- PPO is numerically identical across all four dimensions down to the exported summary precision. Under the current quick-profile training budget, richer state does not change its learned outcome materially.
- SAC remains negative in every dimension and becomes especially poor at 50D.

This is strong evidence that dimensionality is not an isolated “more is better” dial. It interacts with optimization method and training budget. The added state dimensions are useful enough to improve the best RL frontier, but they do not produce universal improvement across all controllers.

12.4 Cross-dimension figures

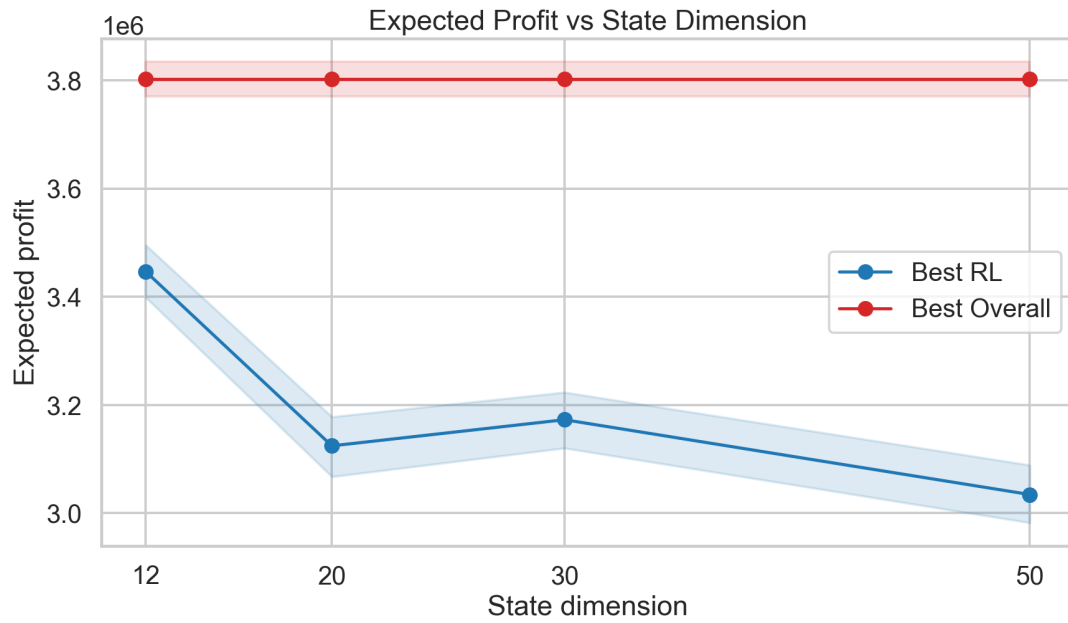


Figure 6: Profit versus state dimension for the best RL controller and the best overall controller.

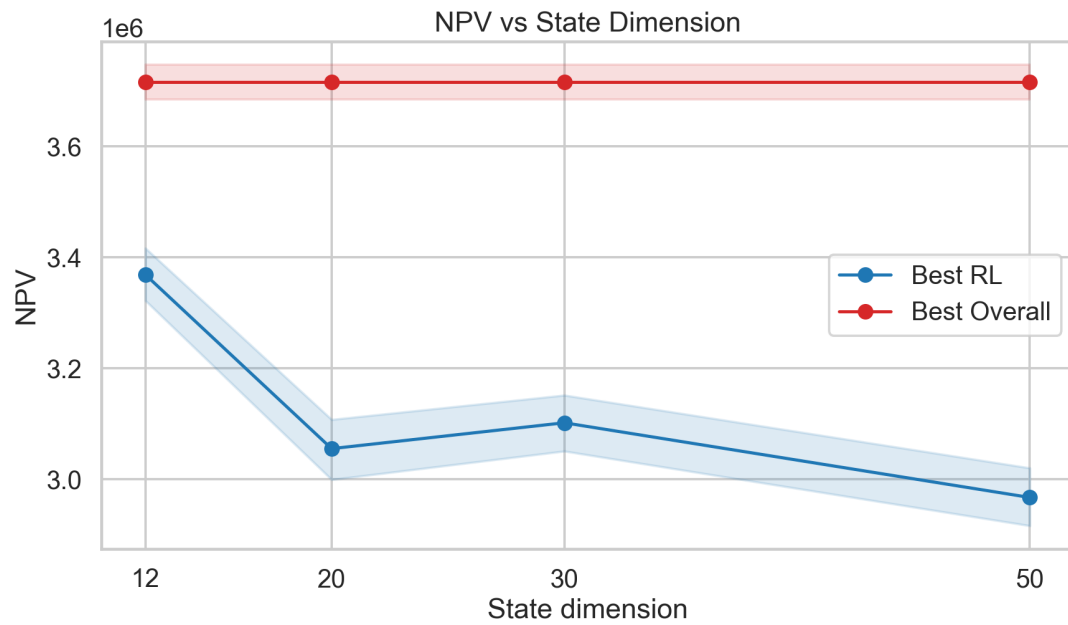


Figure 7: NPV versus state dimension.

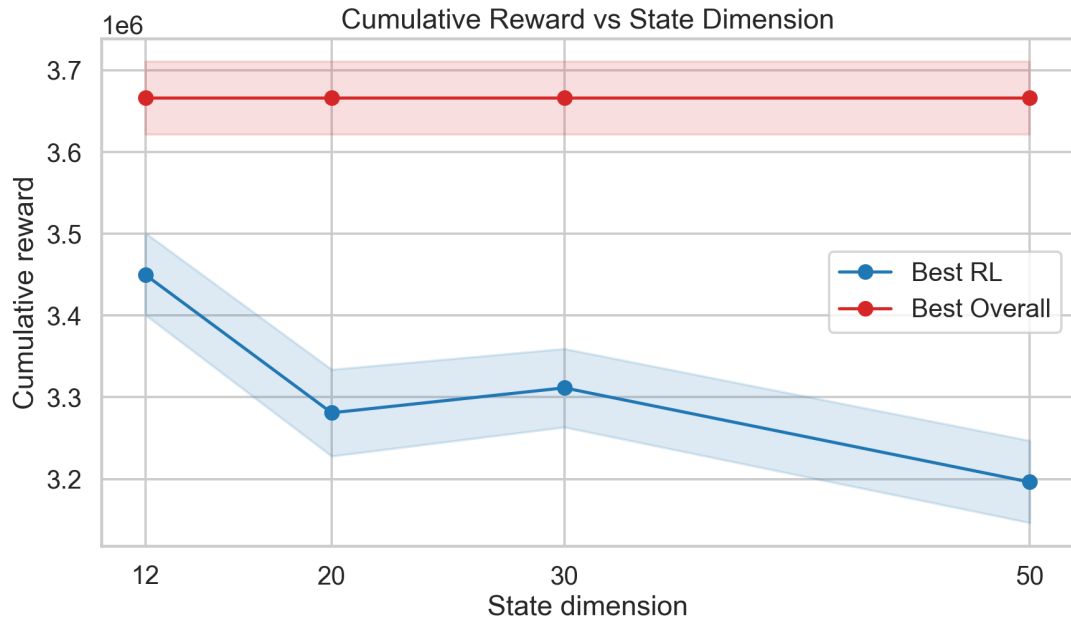


Figure 8: Cumulative reward versus state dimension.

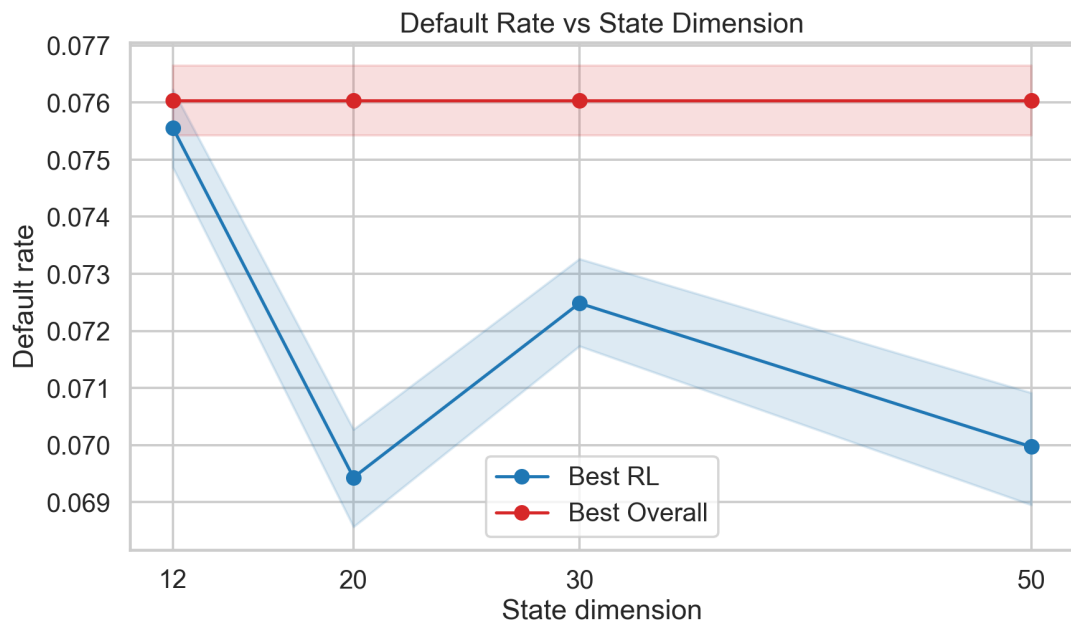


Figure 9: Default rate versus state dimension.

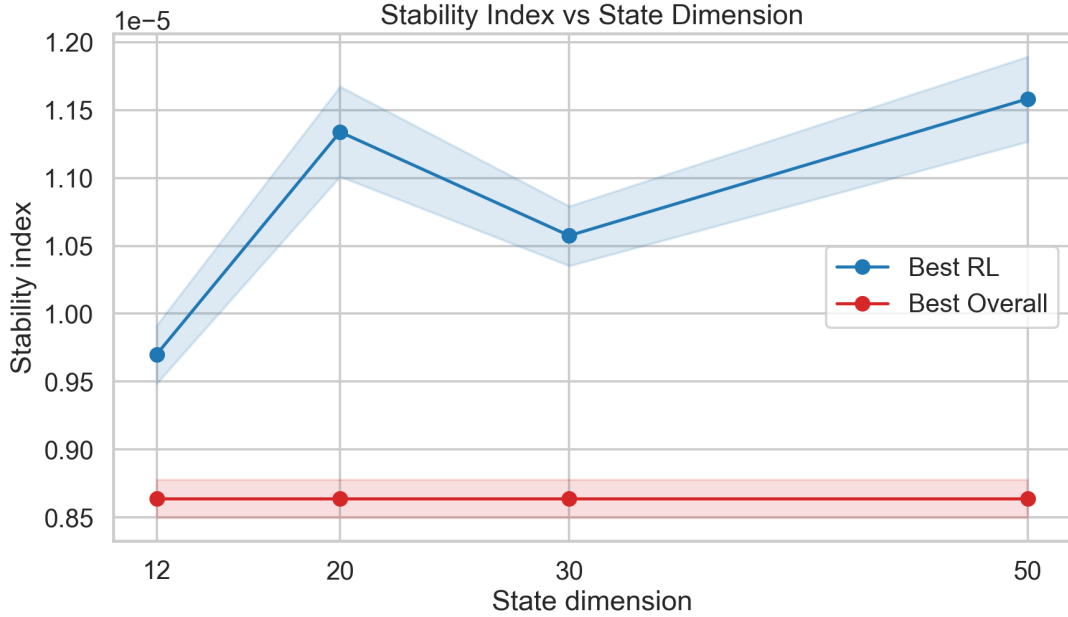


Figure 10: Stability index versus state dimension.

Figures 6–10 summarize the same story visually. The best-overall line is flat because the best baseline does not depend on the varying state vector. The best-RL line rises from 12D to 30D on profit, NPV, and reward, then flattens. Default rate improves sharply at 30D, then drifts slightly upward at 50D. Stability changes little, which means the main gain is not a large volatility reduction but a better balance of selectivity, realized performance, and capital usage.

13 Scenario-Level Results and Interpretation

13.1 Why scenario-level analysis is necessary

The overall leaderboard is informative but incomplete. Because the reward aggregates across six scenarios, it can hide the situations in which RL already matches or exceeds the best non-RL policy and the situations in which it still falls far short. The repository exports scenario-level summaries precisely to avoid that loss of structure.

Two different notions of “winner” must be distinguished.

- The *reward-based winner* is the controller selected by the repository’s official ranking logic.
- The *highest-profit winner* is the controller with the largest `expected_profit_mean` in the output tables.

These are not always the same controller because the reward function is intentionally multi-objective.

13.2 Scenario trade-offs in the richest state

Table 8 focuses on 50D because it is the richest observation design and makes the reward-profit-risk distinction especially visible.

Scenario	Reward-based winner	Highest-profit winner	Lowest-default winner
adverse_stress	static_threshold	constraint_aware_weekly	risk_aware_weekly
base_market	profit_oriented	profit_oriented	risk_aware_weekly
class_imbalance_shift	constraint_aware_weekly	profit_oriented	risk_aware_weekly
drift	profit_oriented	profit_oriented	risk_aware_weekly
noise	profit_oriented	profit_oriented	risk_aware_weekly
split_policy_dynamics	constraint_aware_weekly	profit_oriented	risk_aware_weekly

Table 8: Reward-profit-risk leaders in the 50D run, computed from `outputs/dim_50/main_scenario_summary.csv`.

This table shows why the repository cannot be summarized by one sentence such as “baseline wins” or “RL wins.” The answer depends on the metric and the scenario.

13.3 Adverse stress

Adverse stress remains the hardest scenario for RL in every observation size. The reward-based winner is always `profit_oriented`, with profit 29,336.42, reward 25,150.74, and default rate 0.1178. The best RL result is much weaker: in 12D, the best RL controller is `double_dqn` with negative profit. In 30D, DQN recovers to a small positive profit of 3,413.22 but still trails the baseline heavily. In 50D, DQN falls back to $-12,023.13$.

This pattern is plausible given the scenario design. Adverse stress combines a shock, stronger default pressure, higher noise, and larger loan amounts. Under those conditions, the cost of over-acceptance rises sharply. The myopic-looking `profit_oriented` baseline is not actually reckless here because the current-cohort profit preview already reflects much worse cohort economics when the market deteriorates. Its weekly grid search therefore reacts by becoming highly selective, and that conservatism is rewarded.

The cross-week snapshots reinforce the same interpretation. In 30D adverse stress, DQN improves substantially over its 12D and 20D versions, but at week 20 cumulative profit remains only about 7,482.68 compared with `profit_oriented`’s 22,046.88. Richer state helps the RL policy survive stress more gracefully, yet it does not close the gap to the best baseline within the current quick-profile budget.

13.4 Base market

Base market is where RL looks strongest. In 12D, DQN is already the reward-based winner with profit 86,519.38 and reward 101,967.98. In 20D, Double-DQN becomes the scenario winner and pushes profit to 94,321.51. In 30D, Double-DQN still leads the scenario on reward, while in 50D DQN becomes the clear leader with profit 92,105.55, reward 110,521.41, and default rate 0.1134.

This result is economically sensible. Base market offers the cleanest environment for representation value to matter. The controller does not have to fight a regime break as severe as adverse stress, so richer state can be used to set a more efficient selective policy. The 50D DQN result is notable because it exceeds the `profit_oriented` baseline not only on reward but also on raw profit and default rate. That is the clearest case in the exported results where RL converts richer portfolio state into both higher return and better credit quality.

At the same time, the base-market result should not be over-generalized. The winning RL agent changes across dimensions, and Double-DQN collapses at 50D overall despite once leading this scenario. This again shows that scenario-level success and global robustness are not the same thing.

13.5 Class-imbalance shift

Class-imbalance shift is the most consistent win for the conservative rule-based benchmark. The reward-based and profit-based winner is always `constraint_aware_weekly`, with profit 70,649.29 and reward 81,420.58. This makes intuitive sense because the scenario increases the share of lower-quality new applicants. A controller that explicitly internalizes approval, default, and capital penalties has a structural advantage.

The RL story is still important, however. The best RL profit rises from 56,746.75 at 12D to 65,715.59 at 50D, and the lowest default rate in 30D and 50D belongs to DQN rather than to the constraint-aware baseline. This indicates that richer state helps RL recognize composition risk and respond more selectively. What it does not yet do in the quick profile is beat the explicit rule search that is already well aligned with the scenario.

13.6 Drift

In the drift scenario, `profit_oriented` is the reward-based winner throughout the dimensionality sweep, with profit 59,526.10 and reward 68,132.12. The best RL policy improves from around 40,685.98 in 12D to 52,005.69 in 30D, then softens to 46,567.72 in 50D.

This is a useful counterexample to a naive “more dimensions always help” claim. The 30D state, which adds lagged and rolling history, is especially well suited to slow deterioration because drift is a temporal phenomenon. The 50D state adds more descriptors, but those extra coordinates do not improve DQN further in this scenario. The likely implication is that the 30D historical layer captures the essential signal, while the 50D enlargement adds modeling burden without equivalent incremental value.

13.7 Noise

The noise scenario reveals the difference between profit maximization and shaped reward most clearly. The highest raw profit belongs to `risk_aware_weekly` at 97,377.95, while the reward-based winner remains `profit_oriented` at profit 94,589.22 and reward 112,586.40. In 30D and 50D, DQN achieves lower default rates than both of those baselines and still posts very strong profits, namely 88,596.00 in 30D and 89,396.58 in 50D.

The interpretation is nuanced. Under noisy weekly volumes and measurement disturbance, the risk-aware baseline appears to exploit current profit well, but the reward function still values the more balanced profile produced by `profit_oriented`. At the same time, richer-state DQN becomes meaningfully safer, with default rates around 0.1133 in 30D and 0.1160 in 50D, compared with `risk_aware_weekly`’s 0.1222. This is one of the best examples of portfolio-quality trade-off in the current outputs: RL does not win the main ranking, but it creates a more conservative portfolio without collapsing profit.

13.8 Split-policy dynamics

The split-policy scenario is designed to reward controllers that can benefit from separate thresholds. The strongest reward-based controller remains `profit_oriented` across all dimensions, but the gap to the best RL agent shrinks sharply as state richness increases. The best RL profits are 114,185.42 at 12D, 117,912.12 at 20D, 112,439.04 at 30D, and 119,918.57 at 50D.

The most important observation is not only that 50D DQN reaches almost the same profit as the baseline. It also achieves the lowest default rate in the 50D scenario summary at approximately 0.1093, compared with `profit_oriented`'s 0.1112. This is exactly the kind of pattern the richer segmented state was meant to uncover: when new and repeat segments evolve in opposite directions, a controller with better segment-specific state can approach baseline-level profitability while becoming slightly safer.

14 Threshold Behavior and Dynamic Response

14.1 What threshold volatility actually shows

At first glance, the main summaries may suggest that the best RL policies are extremely stable because their reported `threshold_volatility_mean` is exactly zero. That interpretation is correct, but only in a specific sense. Threshold volatility is defined from week-to-week changes within an episode. A value of zero therefore means that the policy keeps the same threshold pair throughout the episode.

The exported weekly logs confirm that this is exactly what happens in the quick-profile winners. For the best RL controller in each dimension, 100% of evaluation episodes are constant-threshold episodes. The improvement from 12D to 30D does not come from more active week-to-week motion. It comes from selecting better threshold pairs.

14.2 Episode-static RL winners

The best RL controllers in the quick profile choose only a small set of episode-level threshold pairs:

- 12D DQN selects the pairs (65, 45), (85, 35), and (85, 50) across episodes.
- 20D Double-DQN selects (65, 40), (75, 50), and (80, 55).
- 30D DQN selects (70, 65), (85, 35), and (85, 50).
- 50D DQN selects (70, 45), (85, 40), and (85, 45).

This fact is important for honest interpretation. In the active quick-profile experiment, richer state helps the controller identify a better static split policy for the episode more than it induces genuinely adaptive weekly threshold movement. That does not invalidate the dimensionality result. It simply narrows its meaning. The current evidence supports “better state improves policy quality” more strongly than “better state produces richer within-episode adaptation.”

This also helps explain why the leading dynamic baselines still perform so well. Their threshold-volatility means are large—about 9.83 for `profit_oriented` and 9.60 for `constraint_aware_weekly`

in the overall summaries—because they reoptimize or adjust thresholds every week. The quick-profile RL winners, by contrast, are strong mostly as episode-level selectors of static threshold pairs.

14.3 Early-loss / late-gain diagnostic

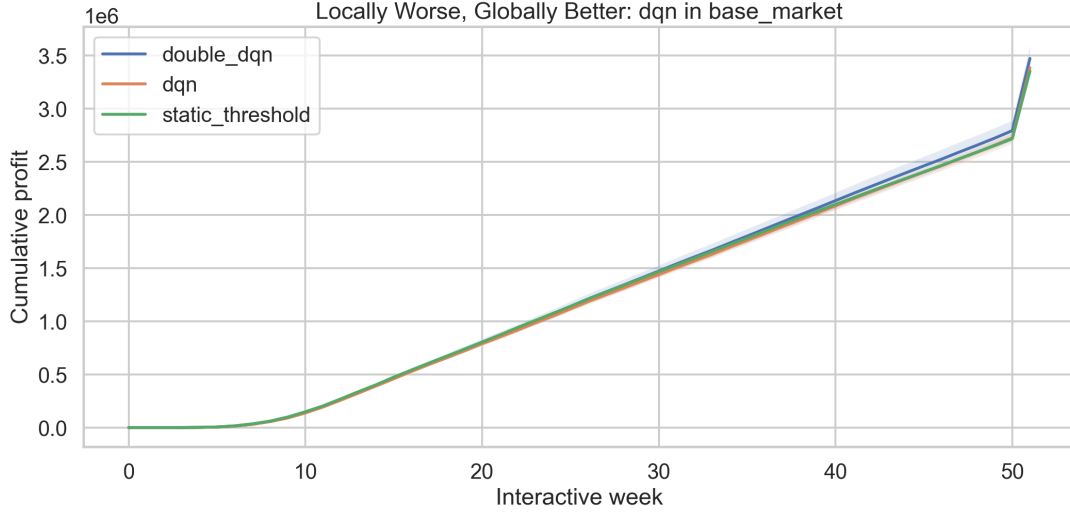


Figure 11: Diagnostic example of a locally worse but globally better trajectory from the 30D output folder.

Figure 11 is useful conceptually even though it is only a diagnostic. It illustrates why delayed cash-flow accounting matters for thesis interpretation. A policy can fall behind early because stricter acceptance reduces immediate cohort contribution, then recover strongly once weaker loans that would have defaulted are never originated. That temporal reversal is exactly the type of behavior that a weekly delayed-outcome simulator is designed to capture and that static one-step classification metrics would miss.

15 Figures for Dimension Comparison

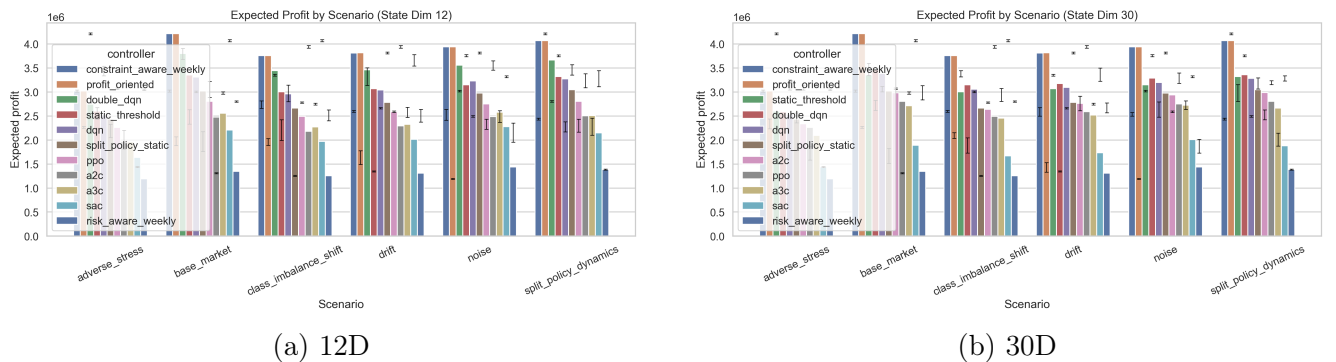
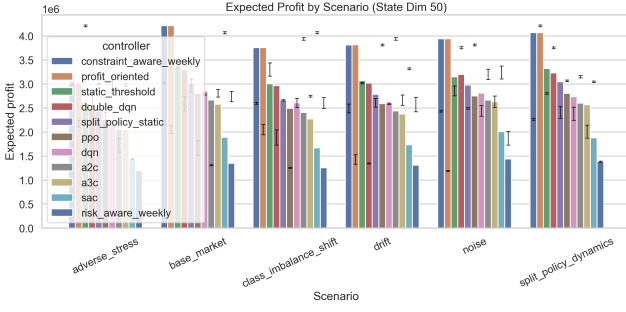
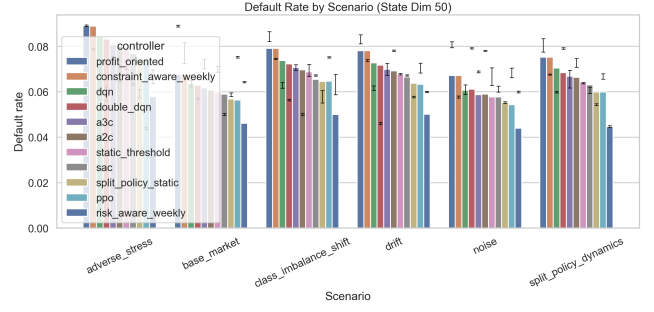


Figure 12: Scenario-level profit comparison for two representative state sizes.

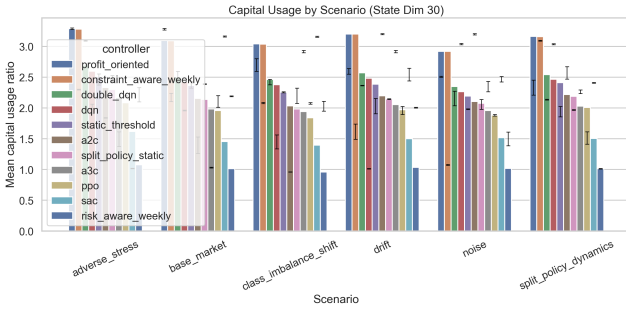


(a) 50D

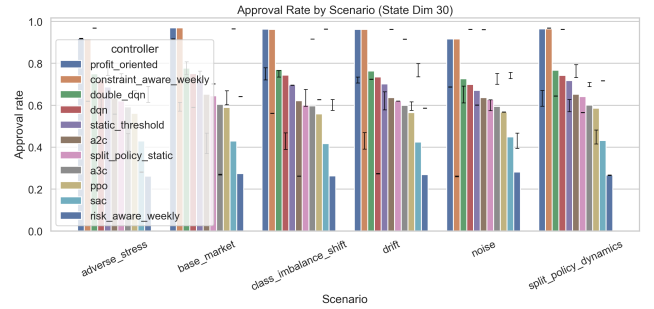


(b) 50D default-rate comparison

Figure 13: Rich-state scenario view of profit and default trade-offs.



(a) 30D capital usage



(b) 30D approval rate

Figure 14: Portfolio-allocation trade-offs for the 30D run.

Taken together, Figures 12–14 show why the 30D and 50D results are best understood through scenario structure rather than through one scalar curve only. The richest state does help in several places, especially `base_market`, `class_imbalance_shift`, and `split_policy_dynamics`, but it does not remove the RL deficit under `adverse_stress` and does not overturn the dominance of conservative rule-based search in `drift`.

16 Discussion

16.1 Why 30D is the most balanced state size

The 30D state is the strongest compromise in the current outputs. It delivers the largest jump in best-RL reward, improves profit and NPV materially, lowers default rate relative to 20D, and does so with lower approval and capital usage. In other words, it improves the quality of selectivity rather than only the quantity of lending.

The extra information added between 20D and 30D is not arbitrary. It is precisely the historical memory layer: lagged approval, lagged realized default, lagged realized profit, lagged capital usage, rolling means, rolling profit variability, and lagged threshold changes. Those variables are well matched to a weekly control problem because they tell the controller whether the latest week is exceptional or part of an emerging pattern. That is especially useful in drift, noise, and composition-shift environments.

16.2 Why 50D saturates

The move from 30D to 50D still adds meaningful descriptors, but their incremental value is smaller. Segment-level means, rolling reward moments, cumulative totals to date, and threshold-gap history make the observation richer, yet the best RL reward no longer improves. In practice, this looks like a saturation frontier: once the controller already knows current performance, recent history, and lagged threshold behavior, the additional segmentation and stability coordinates do not produce a comparable gain in decision quality under the current training budget.

There is also a more cautionary reading. The 50D run is not merely “no better.” It actively harms some algorithms. Double-DQN falls from a positive reward of 69,812.02 at 20D to $-79,329.16$ at 50D, and SAC deteriorates sharply as well. This means that the 50D state can be viewed as both richer and more optimization-demanding. The state itself is not bad, but its control value is conditional on the learning algorithm and the training budget.

16.3 Why the best baseline still wins overall

The persistent dominance of `profit_oriented` and `constraint_aware_weekly` is not accidental. Both baselines exploit privileged structural alignment with the environment.

- They perform explicit weekly search over the same threshold grid that the environment exposes.
- They do not have to learn a value function or policy representation from limited episodes.
- Their decisions are directly tied to the simulator’s current weekly economics and constraint structure.

Under the quick profile, RL has only 12 training episodes per seed, which is deliberately small. The fair interpretation is therefore not “RL failed.” The fair interpretation is that strong hand-crafted policy search remains extremely competitive in a well-structured simulator, while RL becomes meaningfully better when the state is informative enough and the scenario is favorable enough.

16.4 Portfolio-quality trade-offs

The dimensionality experiment is most interesting where several metrics move in opposite directions. The best RL controller at 20D increases approval rate and capital usage relative to 12D, but also slightly worsens default rate. The 30D controller reverses that pattern: it becomes more selective, uses less capital, and improves reward strongly. The 50D controller then relaxes again, with higher approval and capital usage but nearly flat reward.

This sequence is exactly the kind of portfolio trade-off that a thesis should emphasize. Better results do not always mean “approve more” or “reduce default” in isolation. What matters is the joint movement of profit, default, capital usage, and stability. The 30D result is strong because the joint movement is favorable. The 50D result is weaker because the additional approval and capital consumption do not translate into additional reward.

16.5 What the current results do and do not establish

The current evidence establishes several limited but useful findings.

- State design matters materially for RL quality.
- The marginal value of additional state information is non-linear.
- Richer state can improve profitability while preserving or even improving risk metrics in some scenarios.
- The benefit of richer state is algorithm-dependent.

It does *not* establish that 50D is universally too large, that 30D would remain optimal under a much larger training budget, or that RL is incapable of learning genuinely dynamic weekly policies. Those stronger claims would require a longer-horizon training study, not only the current quick-profile outputs.

17 Limitations

Several limitations should be stated directly.

- The simulator is synthetic. Its segment parameters and scenarios are economically plausible, but they are not calibrated to confidential production portfolios.
- The quick profile is intentionally small. Twelve training episodes per seed are sufficient for an end-to-end controlled experiment, not for a final convergence claim.
- The reward is a designed objective, not a discovered one. The relative ranking of controllers is therefore partly a function of the chosen trade-off between profitability, approval pressure, capital usage, and volatility.
- The exported confidence intervals quantify uncertainty inside the simulation generator. They do not, by themselves, establish external validity on real bank data.
- The best RL policies in this profile are episode-static threshold pairs. Therefore, the current evidence for adaptive weekly threshold movement is weaker than the evidence for better episode-level threshold selection.

These limitations do not weaken the core methodological contribution. They simply define the correct boundary of the claim.

18 Research Contribution

The contribution of the study lies less in inventing a brand-new RL algorithm and more in combining several research ingredients in one controlled experimental design.

- **Methodological contribution:** credit-scoring control is formulated as weekly portfolio threshold management rather than one-shot application classification.

- **Simulation contribution:** delayed outcomes, late recovery, realized NPV, capital usage, and terminal settlement are made explicit parts of the environment.
- **Experimental contribution:** new and repeat applicants are controlled through separate thresholds, and state representation is varied through a controlled 12D–20D–30D–50D experiment.
- **Benchmarking contribution:** learned policies are evaluated against a diverse and economically interpretable baseline set rather than against a weak static comparator only.
- **Empirical contribution:** the results identify where RL works, where it is limited, and why scenario-level robustness is more informative than a single average score.

This is why the thesis contribution should be described as a contribution in *problem formulation, simulation design, experimental methodology, and robustness interpretation*, not only as a leaderboard comparison.

19 Conclusion

The goal of this thesis was to evaluate the robustness of a reinforcement learning framework for credit-scoring threshold optimization under different state representations, market scenarios, and portfolio constraints. The research question asked under which conditions reinforcement learning provides a robust improvement over rule-based threshold policies in a delayed-outcome credit-scoring environment.

The thesis achieved this goal by developing the original threshold-control idea into a richer experimental framework. The implemented environment models weekly split thresholds for new and repeat customers, delayed repayments and defaults, late recoveries, realized profit and NPV, capital usage, reward penalties, and terminal settlement. The evaluation covers four state representations, six market scenarios, four RL algorithms, and five rule-based or static baselines. Statistical evidence was added where run-level data allowed paired comparisons.

The first hypothesis, concerning state representation, receives only partial support. The paired reward comparison between 30D and 12D DQN is not statistically significant ($p = 0.41$), meaning the additional state information does not reliably improve the best RL frontier within the full training budget. However, 30D is significantly better than 50D ($p \approx 0$), confirming that the 50D segmentation layer introduces diminishing returns. The 30D state therefore remains the most balanced representation in the sense that it avoids the degradation observed at 50D, even if it does not clearly surpass 12D.

The second hypothesis is strongly confirmed. Across all four dimensions, RL controllers win zero of six scenarios by cumulative reward, zero by expected profit, and zero by lowest default rate. The best overall controller is `constraint_aware_weekly` (profit 3,802,547, reward 3,666,052), essentially tied with `profit_oriented`. RL controllers are not irrelevant: several produce lower default rates than the baselines in specific scenarios, and Double-DQN is the strongest RL agent across all dimensions. But the evidence against universal RL superiority is unambiguous.

The third hypothesis is rejected for the full-profile setting. In the full-profile adverse-stress run, 30D DQN is significantly worse than `static_threshold` at the final horizon ($-41,507$ profit, $p = 2.8 \times 10^{-7}$). The “locally worse, globally better” pattern observed in the quick profile does not

generalise to the full training protocol. Full-horizon evaluation with terminal settlement remains methodologically important, but in this experiment it reveals that the static baseline is persistently stronger than the DQN policy under adverse stress.

Overall, the main answer to the research question is conditional. Reinforcement learning can be useful for dynamic credit-scoring threshold optimization when the state representation contains enough portfolio history and when the market scenario contains exploitable structure. However, strong rule-based baselines remain difficult to beat, especially under stress and drift, and more features do not necessarily lead to better policies. The project should therefore be interpreted as a robustness evaluation of RL-based threshold control, not as a claim of unconditional RL superiority.

The main limitations remain the synthetic nature of the simulator, the short quick-profile training budget, the designed reward function, and the episode-static behavior of the best RL policies in the current runs. Future work should test longer training profiles, reward-sensitivity analysis, stronger adaptive RL policies, calibration on real or semi-synthetic credit data, and more explicit regulatory or capital constraints.

Statement on the Use of AI Tools

During the preparation of this thesis, large language model assistants (Claude by Anthropic and ChatGPT by OpenAI) were used as auxiliary tools to support literature search, grammar checking of selected passages, and explanation of mathematical and algorithmic concepts. All experimental design, simulator implementation, data analysis, interpretation of results, and academic responsibility remain entirely with the author.

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